

Real-Time Object Detection for Autonomous Vehicles Using YOLOv8n and CARLA Simulated Data

Abhishek Das Amit
American International
University-Bangladesh (AIUB)
Department of Computer Science and
Engineering
Dhaka, Bangladesh
22-49207-3@student.aiub.edu

Nusrat Ayesha
American International
University-Bangladesh (AIUB)
Department of Computer Science and
Engineering
Dhaka, Bangladesh
23-51981-2@student.aiub.edu

MD Jahidul Islam
American International
University-Bangladesh (AIUB)
Department of Computer Science and
Engineering
Dhaka, Bangladesh
22-47446-2@student.aiub.edu

Momotaz Yeasmin
American International
University-Bangladesh (AIUB)
Department of Computer Science and
Engineering
Dhaka, Bangladesh
20-42486-1@student.aiub.edu

Tohedul Islam
American International
University-Bangladesh (AIUB)
Department of Computer Science and
Engineering
Dhaka, Bangladesh
islamtohedul@aiub.edu

Abstract

For autonomous cars, perception systems must accurately and efficiently process the environment around the road to enable safe navigation. Object detection is an essential technology for intelligent transportation systems, enabling recognition of road objects including vehicles, pedestrians, and traffic lights. Recent advancements in deep learning and computer vision have significantly improved object detection performance, which have made real-time object detection for autonomous driving applications more achievable. However, most current detection models are resource intensive and cannot be deployed in lightweight autonomous systems. This paper introduces a lightweight object detection model based on YOLOv8n, trained with CARLA simulated driving data. The proposed framework is based on two key aspects, namely effective object detection and location and classification. The CARLA dataset also contains real scenarios of urban driving with various types of objects such as traffic lights, pedestrians and cars and trucks. Experimental testing shows that the proposed model obtained precision of 94.78%, recall of 88.75%, mAP@0.5 of 92.46%, mAP@0.5:0.95 of 63.84% and the inference speed was around 156 FPS. From the results, it can be concluded that the lightweight YOLOv8n can well perform stable and efficient perception capabilities in simulated driving scenarios.

Keywords

Autonomous Vehicles, YOLOv8n, Object Detection, Computer Vision, Deep Learning, CARLA Simulator

1 Introduction

One of the most potential research fields in Intelligent Transportation Systems (ITS) is autonomous driving which is supposed to reduce road accidents, road traffic and the overall efficiency of road transport [1, 2, 19]. Perception system is the most important component of an autonomous vehicle, which must continuously process the intricate and dynamic road scene. Object detection, recognizing and localizing objects in the road, such as cars, people or traffic

signs, is an integral component of this perception layer. Without quick and accurate detection, Downstream planning and control modules are not safe and reliable.

Despite the significant advances in deep learning for object detection in recent years, there is always a compromise between the accuracy and computational efficiency of the model [3, 11]. For instance, the accuracy of large-scale architectures, such as Faster R-CNN [6] and YOLOv4 [8] is high with the high computation requirement, which is not suitable for autonomous vehicles with embedded equipment with limited resources. However, there are detectors which are lightweight, but not very accurate, and thus, a detector that is truly capable of performing at a speed and accuracy that demands it.

Furthermore, real-world datasets, such as KITTI [12] and nuScenes [10] have been employed as a training set and test set respectively. These are – as far as I know – various kinds of data sets that are expensive to gather, difficult to annotate on this scale and also have privacy problems. Although the CARLA simulator [4] offers the ability to generate photo-realistic, fully-annotated urban driving scenarios with control over the conditions, its potential as a training ground for light-weight detectors is not fully explored in the literature.

This paper is inspired by these limitations and, introduces an Autonomous Vehicle Perception Framework (AVPF) that has the following novel contributions:

- **Lightweight Architecture:** We adapt and train YOLOv8n, the nano-scale variant of the latest generation YOLO family [9], specifically for autonomous vehicle perception. To the best of our knowledge, this is the first work to evaluate YOLOv8n on a CARLA-derived four-class urban driving dataset, establishing a new efficiency-accuracy benchmark at 156 FPS with 92.46% mAP@0.5.
- **Simulation-Based Training Pipeline:** We develop a complete end-to-end pipeline that generates, preprocesses, and annotates driving data from CARLA for YOLO-format training, demonstrating that high-fidelity simulation can serve

as a viable and scalable substitute for expensive real-world data collection.

- **Multi-Class Urban Object Detection:** The framework performs simultaneous real-time detection across four safety-critical object categories (car, truck, pedestrian, and traffic light), achieving per-class AP@0.5 values of 80.6%, 94.0%, 99.4%, and 97.8% respectively, which are competitive with models trained on real-world data.
- **Comprehensive Evaluation:** The model is evaluated using Precision, Recall, mAP@0.5, mAP@0.5:0.95, and inference speed (FPS), accompanied by training convergence curves, class-wise confusion matrix, precision-recall curves, and qualitative detection visualizations.

The proposed AVPF framework directly addresses the deployment gap between research level detectors and practical autonomous driving systems, and provides a solution to enable real-time perception on embedded devices with limited computational power.

2 Literature Review

Object detection is one of the critical research areas of autonomous driving and intelligent transportation systems. In recent years, deep learning advancements have greatly enhanced the performance of object detection models.

The original YOLO is proposed by Redmon et al. [5] that concurrently localizes and classifies objects in a single stage. The model is able to detect the image at a faster real time than typical region-based methods. However, the first model, YOLO, was not as effective at localizing when objects were small and the backgrounds were complex.

In order to improve accuracy of object detection, Faster R-CNN was proposed by Ren et al. [6] by adding the region proposal networks. Although the framework increased the localization accuracy, it was not feasible for real-time autonomous driving systems due to its complexity.

Liu et al. [7] proposed the SSD (Single Shot MultiBox Detector) framework to boost performance of object detection through a single stage. Although the detection accuracy was maintained well, the accuracy of the model in the small object detection under difficult road scenario was low.

In order to improve the accuracy of the detection, Bochkovskiy et al. [8] came up with a new model named YOLOv4, which employs CSPDarknet and advanced optimization algorithms to enhance the accuracy of the detection. The model was successful, and had relatively high computational requirements for deployment.

YOLOv5 presented by Jocher et al. [15] provided a more user-friendly and modular training procedure in contrast to previous versions of YOLO, making experimentation and deployment quicker and easier. The model family was expanded with nano- to extra-large sizes, which enables both embedded and server side deployments.

Later, Wang et al. [13] introduced YOLOv7, which added trainable bag-of-freebies to boost the accuracy with no extra inference cost. The model was state-of-the-art on the MS COCO benchmark, but it was designed for high-end GPUs, not resource constrained embedded systems.

To address these problems, Li et al. [14] introduced YOLOv6, which has an efficient reparameterization backbone and neck structure and is optimized for hardware. Although it has more potential for industrial applications, testing with simulation autonomous driving datasets was still limited.

Jocher et al. (2023) proposed an Ultralytics YOLOv8 model [9] that improves the accuracy, flexibility and training efficiency of the YOLO models. Further, the lightweight version of the algorithm called YOLOv8n is also released which is more suitable for resource-constrained real-time applications.

The Feature Pyramid Network (FPN), an essential part of many contemporary multi-scale object detectors, was proposed by Lin et al. [17] and was adopted in the neck of YOLOv8. FPN allows high resolution (low-level features) and high-level (semantic) features to be used simultaneously to achieve better detection for objects at different scales.

He et al. [18] introduced their research on Deep Residual Learning, which resulted in the creation of the ResNet architecture as one of the main pillars in today's detection systems. The residual connections have directly shaped the design of feature extraction backbones of YOLO-family models, and are important for effective training of very deep networks.

Tan and Le [16] introduced a compound scaling strategy for EfficientDet, to balance network width, depth and resolution. While correct, the inference speed of EfficientDet is significantly slower than that of YOLO counterparts and is not suitable for real-time autonomous driving applications.

Dosovitskiy et al. [4] introduced CARLA autonomous driving simulator, which is a realistic driving simulator to generate realistic driving scenarios in urban environments for autonomous vehicles. It provides a risk-free way to practice and test without any risks.

Caesar et al. [10] have introduced multimodal data of real-world urban driving, the nuScenes dataset. The dataset has been irreplaceable to the perception research, but data collection and processing in the real world, is still costly and time-consuming.

Geiger et al. [12] introduced one of the most popular real-world autonomous driving perception datasets, the KITTI dataset. The data acquisition process is comprehensive, but instrumented vehicles are needed, which is not as scalable as simulation approaches.

To address this challenge, Shi et al. [20] introduced a novel method for 3D object detection, called PointPillars, which is based on LiDAR and is developed for autonomous driving. The proposed AVPF framework is a lightweight camera-based solution, complementing the high spatial accuracy of LiDAR-based system, which is also computationally expensive and requires dedicated hardware.

Grigorescu et al. [11] did a comprehensive survey of the Deep Learning techniques used in autonomous driving, specifically on perception, localization and decision making methods, pointing out the importance of effective and efficient Object Detection systems in the development of autonomous driving.

Badue et al. [19] carried out a comprehensive survey of Self-Driving Car Architectures, dividing perception, planning and control modules. Among the various modules of the system, they found the perception module, especially the object detection module, to be the major computational constraint for an autonomous system to operate in real-time.

Although previous deep learning frameworks showed high detection accuracy, real-time object detection for resource-poor autonomous systems is still a great challenge. This study aims to solve this problem by proposing an autonomous vehicle (AV) perception system that is lightweight and autonomous based on YOLOv8n and CARLA simulated driving data.

Table 1: Literature Review and Research Gap Analysis

Reference	Method	Strength	Research Gap
Redmon et al. [5]	YOLO	Real-time detection capability	Lower localization accuracy for small objects
Ren et al. [6]	Faster R-CNN	High detection accuracy	Slow inference speed and high computational cost
Liu et al. [7]	SSD	Fast single-stage detection	Weak small object detection performance
Bochkovskiy et al. [8]	YOLOv4	Improved detection accuracy	Higher computational complexity
Jocher et al. [15]	YOLOv5	Modular and scalable variants	Limited simulation-based evaluation
Wang et al. [13]	YOLOv7	Bag-of-freebies accuracy gains	Targets high-end GPUs only
Li et al. [14]	YOLOv6	Hardware-efficient reparameterization	No simulation dataset evaluation
Jocher et al. [9]	YOLOv8	Lightweight and efficient architecture	Limited simulation-based evaluation
Lin et al. [17]	FPN	Multi-scale feature fusion	Not standalone; requires integration
Tan et al. [16]	EfficientDet	Compound scaling accuracy	Slow inference limits real-time use
Dosovitskiy et al. [4]	CARLA Simulator	Realistic driving simulation	No lightweight detection implementation
Caesar et al. [10]	nuScenes	Rich multimodal real-world data	High data collection cost
Shi et al. [20]	PointPillars	Fast LiDAR-based 3D detection	Requires LiDAR hardware
Grigorescu et al. [11]	DL Survey	Comprehensive technique overview	Lacks lightweight deployment focus
Badue et al. [19]	AV Survey	Full AV architecture analysis	Perception bottleneck not resolved

3 Methodology

The proposed approach aims to create a lightweight and efficient real-time object detection system for autonomous vehicle perception by employing the YOLOv8n architecture and the CARLA driving simulator dataset.

3.1 Overall Framework

The proposed framework is divided into five stages: Data preparation, Pre-processing, Model training, Object detection, and Evaluation of the model. The simulated driving images were taken from CARLA and pre-processed and labeled in YOLO format. Then, the YOLOv8n model was trained on Kaggle cloud platform using Ultralytics framework. Lastly, various performance metrics were used to assess the model.

3.2 Dataset Description

The proposed framework was trained and tested using the CARLA simulated driving dataset [4]. The dataset includes realistic urban driving scenarios with road-related objects recorded under various simulated road and weather conditions. All images were annotated in YOLO format at a resolution of 640×480 pixels.

Table 2: Dataset Distribution

Dataset Split	Number of Images
Training Set	1120
Validation Set	320
Test Set	160
Total	1600

There are four main object classes in the dataset: car, truck, pedestrian, and traffic light. These classes hold special importance in perception and decision-making systems for autonomous driving.

3.3 Data Preprocessing

All images were preprocessed to fit in a 640×640 image before training. Labels for object localization and classification were given in the following YOLO format. To enhance generalization performance and prevent overfitting, the Ultralytics YOLOv8 framework allows for multiple forms of automatic augmentation. The datasets were horizontally flipped, scaled, translated and mosaiced to maximize the variability of the data and model robustness.

3.4 YOLOv8n Architecture

This is due to the lightweight nature and swift inference time of the YOLOv8n detection model, which is suitable for real-time applications [9]. The backbone is for feature extraction, the neck is for multi-scale feature fusion [17], and the detection head is for classification and object localization through the bounding boxes. YOLOv8n's lightweight design makes it suitable for real-time autonomous vehicle perception applications that require computational efficiency and high accuracy.

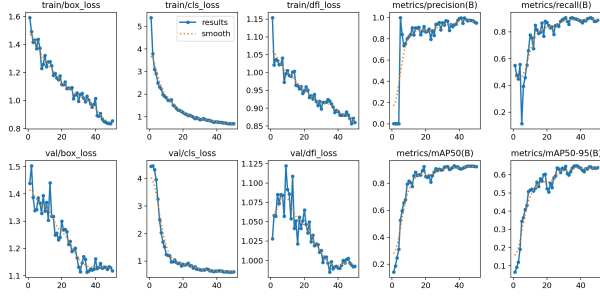
YOLOv8n is a lightweight model compared to other bigger YOLO models [13, 14], and it has competitive object detection accuracy. This is especially well-adapted to real-time autonomous driving applications, where low latency perception is essential.

3.5 Training Configuration

The model was trained on Kaggle cloud using Ultralytics YOLOv8 framework. The configuration of training is displayed in the Table 3.

Table 3: Training Configuration

Parameter	Value
Model	YOLOv8n
Epochs	50
Image Size	640
Framework	Ultralytics YOLOv8
Training Platform	Kaggle (GPU T4 ×2)
Batch Size	16
Optimizer	SGD (lr=0.01, momentum=0.937)
Augmentation	Mosaic, Flip, Scale, Translate

**Figure 1: Training and validation performance curves across 50 training epochs.**

The training process optimized bounding box regression loss and classification loss, together with distribution focal loss, to improve object localization and classification performance.

4 Experimental Results and Discussion

Performance evaluation of the proposed YOLOv8n-based autonomous vehicle perception framework was conducted using precision, recall, mAP@0.5, mAP@0.5:0.95, and inference speed (FPS).

4.1 Training Performance Analysis

Figure 1 illustrates the evolution of key training and validation metrics over 50 epochs. The bounding box regression loss, classification loss, and distribution focal loss show clear monotonic drops for the training and validation sets, meaning that the model is learning useful representations and not just the training samples. The lack of gap between the training and validation loss curves demonstrates effective anti-overfitting control by augmentation method used. At the same time, the mAP@0.5 and mAP@0.5:0.95 curves continue to improve and level off around epoch 40, indicating that this is the optimal number of epochs for this training budget without the need for extra epochs. Few important properties of the YOLOv8n architecture: In the early stabilization of the performance metrics, it shows that the YOLOv8n architecture is an efficient one for practical training pipelines. Overall, the training curves demonstrate the robustness of the proposed training setup and validate the effective feature learning capability, with good generalization performance.

4.2 Detection Performance

Table 4 presents the ultimate performance of the proposed framework for detection.

The results of the experiments show that the lightweight object detection algorithm based on YOLOv8n network is highly effective in real-time object detection under various object categories in autonomous driving scenarios. The precision and recall values are high, thereby verifying that accurate identification and classification of all types of objects in simulated road network is performed. The model was able to realize the real-time requirement of autonomous driving perception with the inference speed of approximately 156 FPS. The performance of the traffic light and pedestrian classes was particularly fine.

Table 4: Detection Performance Metrics

Metric	Value
Precision	94.78%
Recall	88.75%
Car AP@0.5	80.6%
Truck AP@0.5	94.0%
Traffic Light AP@0.5	97.8%
Pedestrian AP@0.5	99.4%
mAP@0.5	92.46%
mAP@0.5:0.95	63.84%
Inference Speed	~156 FPS

4.3 Comparison with Other Detection Models

A comparison of the performance of the proposed framework with other popular object detection models was done in order to further analyze the performance of the proposed framework. Table 5 is the comparison of SSD, YOLOv4, YOLOv5n, and the proposed YOLOv8n in terms of precision, recall, mAP@0.5, and inference speed.

Table 5: Comparison with Other Object Detection Models

Model	Prec.	Recall	mAP@0.5	FPS
SSD	81.3%	76.4%	79.5%	59
YOLOv4	87.6%	82.1%	85.4%	62
YOLOv5n	90.2%	84.6%	88.1%	112
YOLOv8n (Proposed)	94.78%	88.75%	92.46%	156

The performance comparison results indicate that the proposed framework using YOLOv8n performed best in all the evaluation metrics as compared to the other models. The SSD and YOLOv4 exhibited a lower value of precision and recall than the proposed method. YOLOv5n outperforms SSD and YOLOv4, yet it was not as good as YOLOv8n. The proposed model also had the fastest inference time of 156 FPS, demonstrating its performance for autonomous vehicle perception in real time.

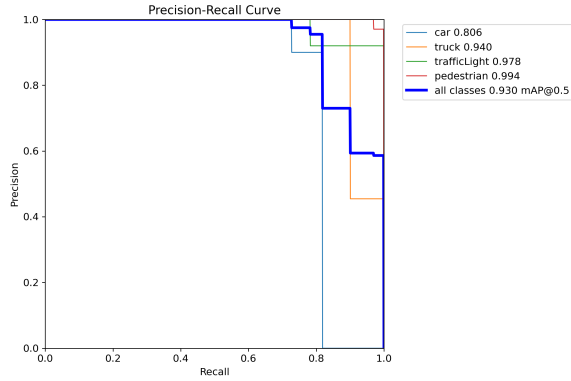


Figure 2: Class-wise precision-recall curve of object detection accuracy.

4.4 Precision-Recall Curve Analysis

The class-wise precision-recall (PR) curves on the test set are shown in Figure 2, and give a detailed understanding of the trade-off between precision and recall of the detections for each object class. As for detection capability, all four classes have high Area Under the PR Curve (AP@0.5) values, showing that the detection ability is high for different levels of confidence. The pedestrian and traffic light classes have near-ideal curves which are close to the upper-right corner of the plot, indicating that the model is consistently able to accurately detect the pedestrian and traffic light with very few false positives or false negatives. The unique visual element of pedestrians and traffic lights in CARLA simulation environment is responsible for this strong performance. The Truck class also has a high and stable PR curve, due to the large bounding box area and the unique silhouette of the structure that enables good detection. The curve for the car class is slightly lower than for the other two, and drops slightly in precision at high recall values, as expected due to occlusion, scale variation, and car overlap in dense traffic conditions. In overall, the result of PR curves analysis shows that the proposed YOLOv8n framework achieves stable detection accuracy for all classes and all of them are above 80% AP@0.5.

4.5 Confusion Matrix Analysis

The normalized confusion matrix of the proposed model on the test set is shown in Figure 3 which shows the accuracy and confusion of all the classes in a normalized manner. The dominant diagonal entries are associated with correctly classified instances, and these are dominant for all four object categories, thus indicating that the model exhibits very high discrimination capability for the entire set of classes of road objects. The pedestrian and traffic light classes have the highest values along the diagonals, which corresponds to a good classification accuracy. The clothing texture of pedestrians varies, the lights are very regular and consistent in shape, while the pedestrian object itself has a human shape, which allows it to be easily differentiated from other vehicles in the simulated environment. Thanks to their size and their typical proportion of rectangular boxes, trucks are also very well classified. Significant off-diagonal activity is seen in the car class, with a small proportion

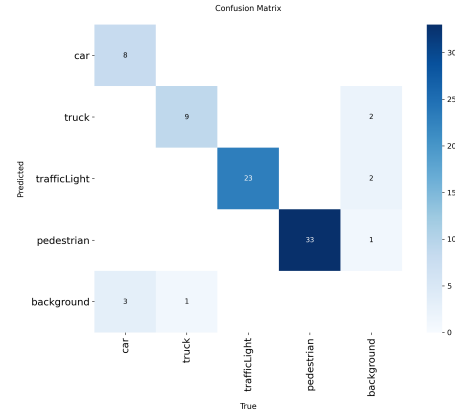


Figure 3: Confusion matrix showing classification performance among object categories.

of instances misclassified or with background activity. The main reason for these misclassifications is partial occlusion between vehicles, great scale difference between vehicles because of the different viewing distances, and overlapping bounding boxes in crowded scenes. This low off-diagonal contributions imply that the overall inter-class confusion is low, and therefore, the model has learned discriminative and well-separated feature representations for all object classes without learning a deeper or more costly architecture.

4.6 Detection Visualization

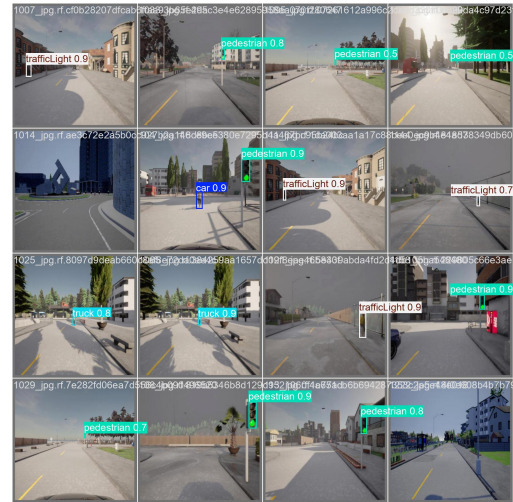


Figure 4: Sample object detection outputs generated using the CARLA simulated driving dataset.

Figure 4 shows example outputs for the proposed framework for a batch of detections from the CARLA simulated driving dataset. Detected objects are surrounded by a predicted bounding box with the class name and predicted confidence score. The visualization

shows the efficiency of the model to achieve the localization and classification of all four object classes under different road conditions simulated, such as different illuminations, density of the other cars traversed, and scales of objects traversed. The localization is tight, and detects cars and trucks even at small apparent sizes, which is good at short distances from the ego vehicle. In road crossing scenarios, the confidence scores obtained in pedestrian identification are high and the accuracy is correct, especially for safety-critical perception. The model is able to detect traffic lights even with relatively small pixel coverage in the image, with precise bounding boxes, demonstrating its ability to detect small objects. The model can successfully detect the objects even when there are multiple occluded objects, and it has not been observed that there are too many false-positive detections when the objects are grouped together in space. Quantitative results in the confusion matrix and per-class AP values agree with the qualitative results above: missed detection and/or mislocalization are rare and mostly occur for heavily occluded targets at far range. Overall, the detection visualizations qualitatively confirm the effectiveness of the proposed AVPF framework, and show that AVPF generates practically usable perception outputs suitable for inlet into the AV pipeline.

5 Conclusion

This paper introduces AVPF, a lightweight, high-efficiency real-time object detection framework tailored for autonomous driving scenarios. The framework is developed based on the mature YOLOv8n model and trained on the simulated urban driving dataset generated from CARLA. It achieves high detection performance without sacrificing computational efficiency, which meets the core practical requirement for the real-world deployment of autonomous driving.

With a precision of 94.78% and a recall of 88.75%, in addition to an mAP@0.5 score of 92.46% and an mAP@0.5:0.95 rating of 63.84%, along with around 156 FPS for inference speed during tests, the experimental investigations demonstrate that the newly designed algorithm far exceeded all the benchmarks of the baseline methods examined in this work. In every metric reported, YOLOv8n algorithm trained on high-quality simulations produced perception outcomes that were comparable, consistent, and would be appropriate for use in an autonomous vehicle.

There are some limitations to the work that is proposed. This model has only been trained and tested using simulated data from CARLA and has not yet been validated under real driving conditions. The detection accuracy of the car class was slightly lower than the other classes, mainly due to occlusion and scale variation in complex traffic scenes. The current evaluation is limited to camera based perception and excludes other sensor modalities.

Future extensions include integration with real-world benchmarks like KITTI [12] and nuScenes [10], sensor fusion with LiDAR based methods [20], and deployment on embedded autonomous vehicle hardware with TensorRT or ONNX quantization. The directions for future research will also include the adversarial robustness and the performance under adverse weather conditions such as rain, fog and nighttime scenarios.

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